

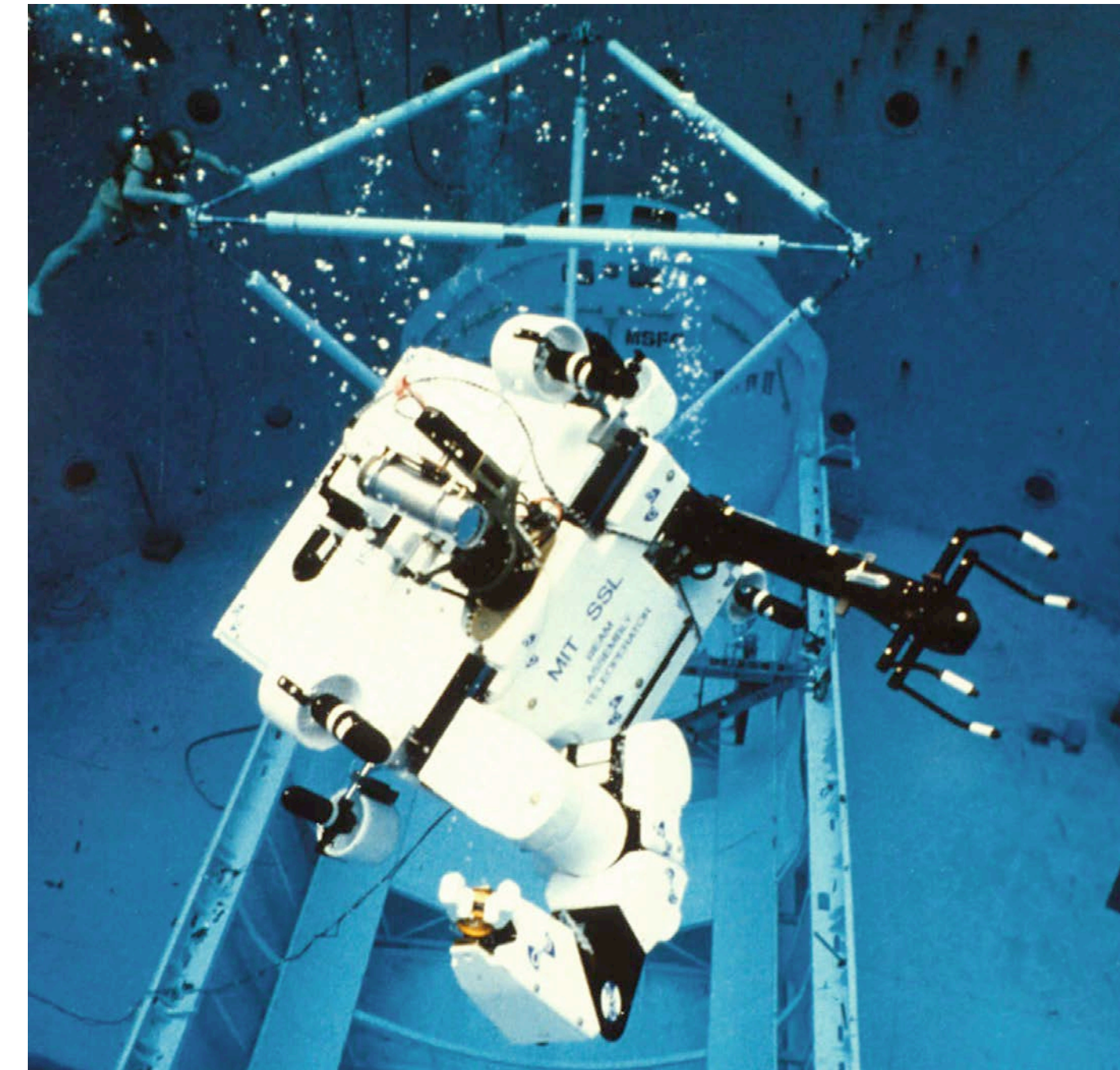
Robotic and Human/Robotic Structural Assembly: Past Experiences, Future Promise

On-orbit EVA Assembly



Based on seven years of neutral buoyancy research, in 1985 the Space Systems Laboratory (then at MIT) flew the Experimental Assembly of Structures in EVA (EASE) structure on STS 61-B. The EASE experimental hardware was designed to maximize dynamic fidelity in neutral buoyancy testing; the flight hardware matched the mass and inertia of the underwater hardware to allow a direct correlation of the simulation data set to flight. This hardware, which was designed to be challenging for a human subject, has since been the primary initial test hardware for all robotic assembly simulations.

Robotic Assembly Operations

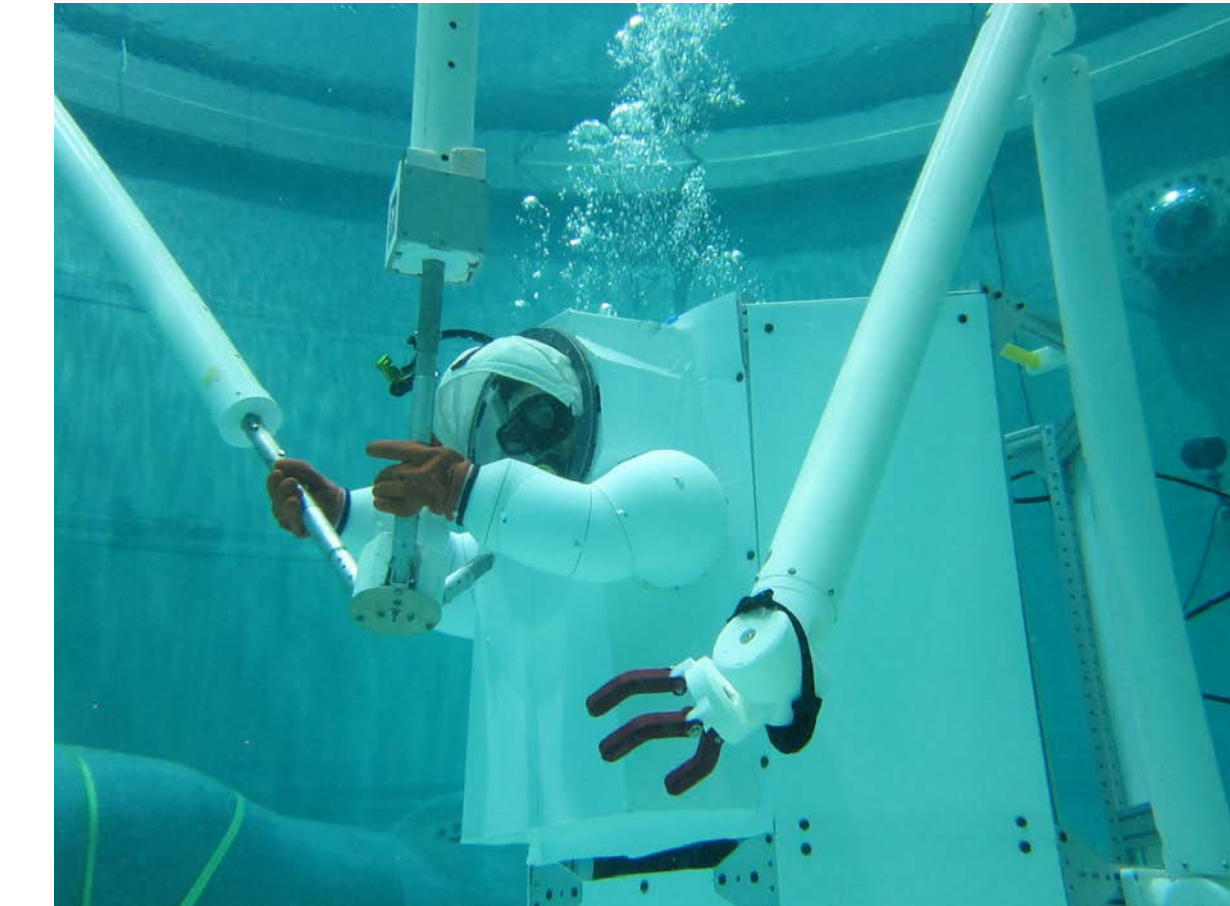


EASE structural assembly

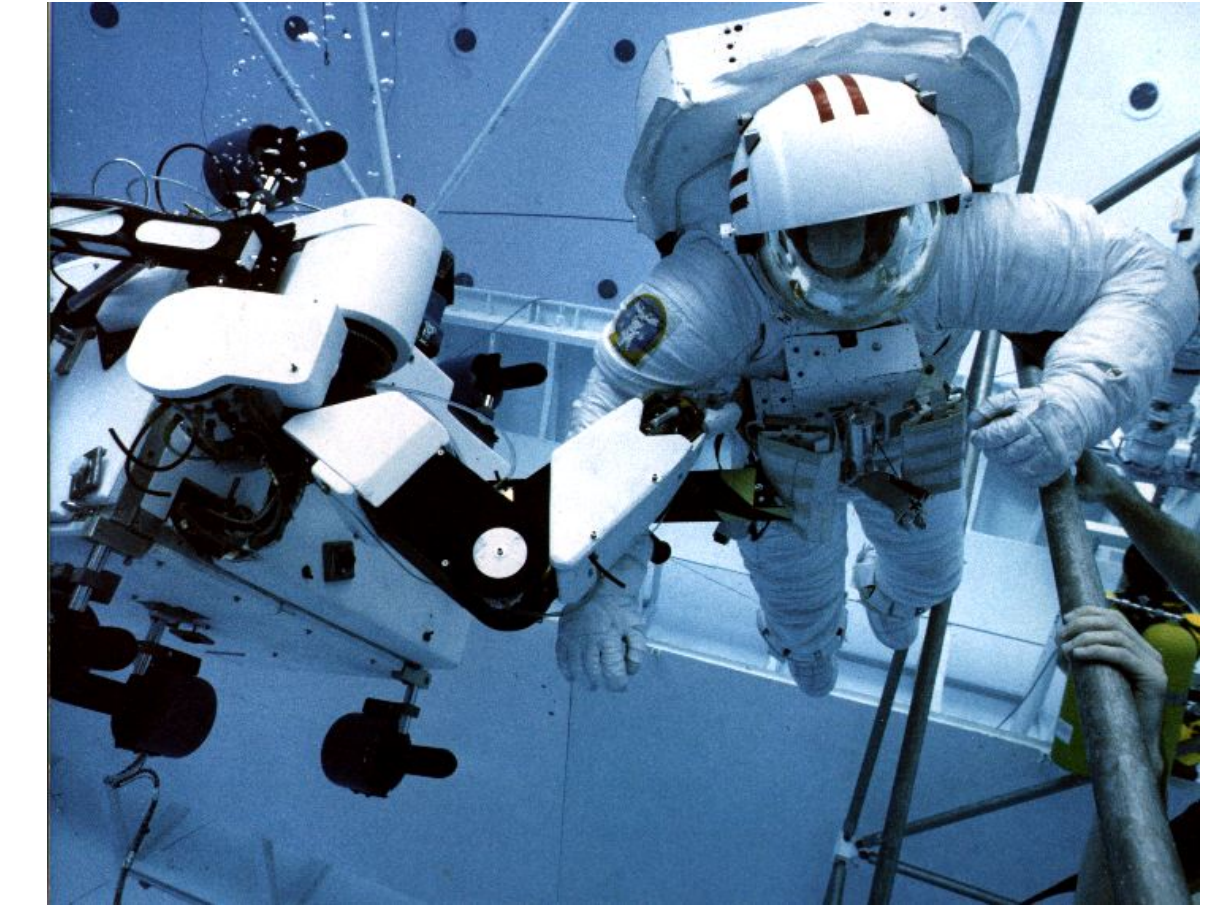


Truss structural assembly

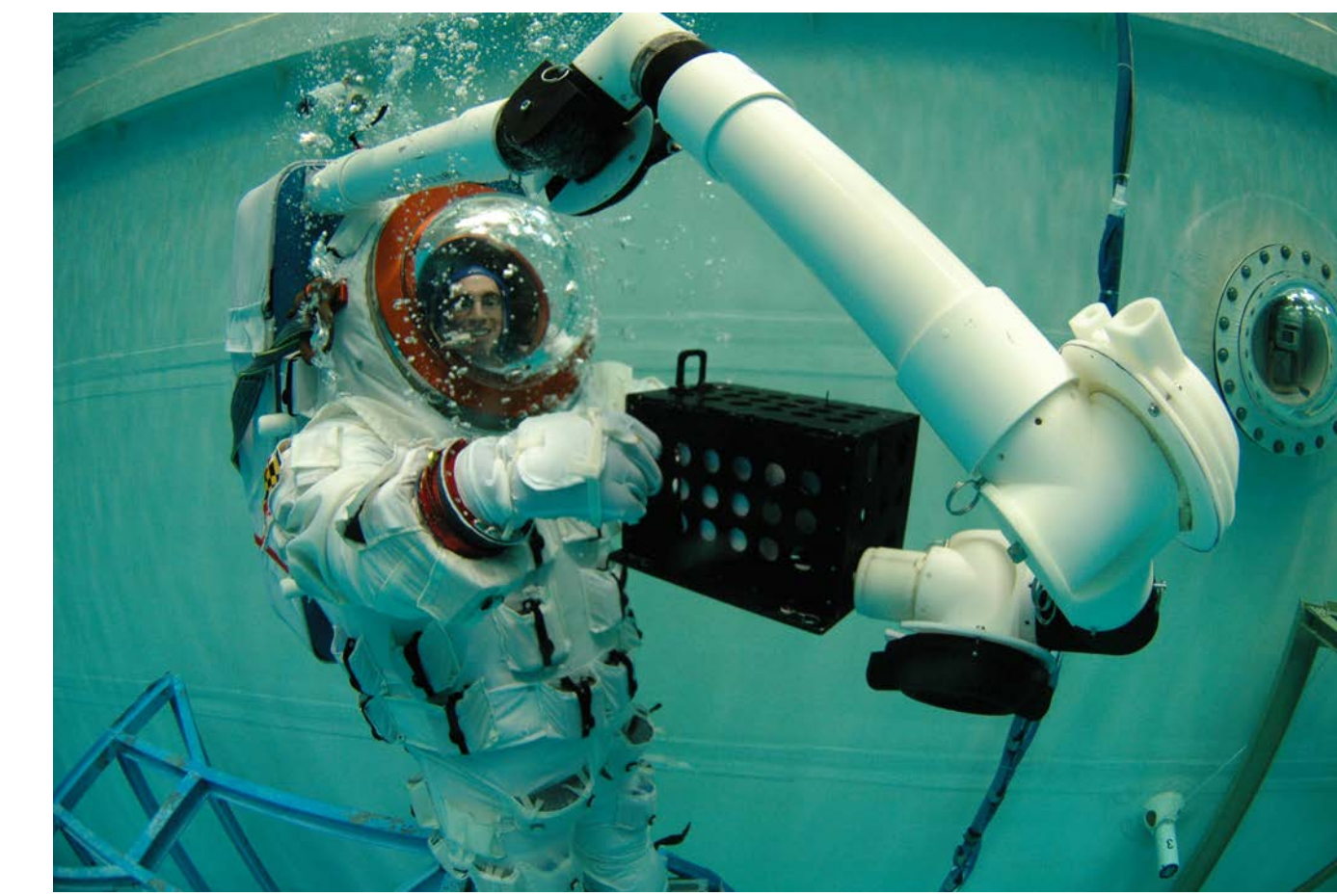
Potential Future Applications



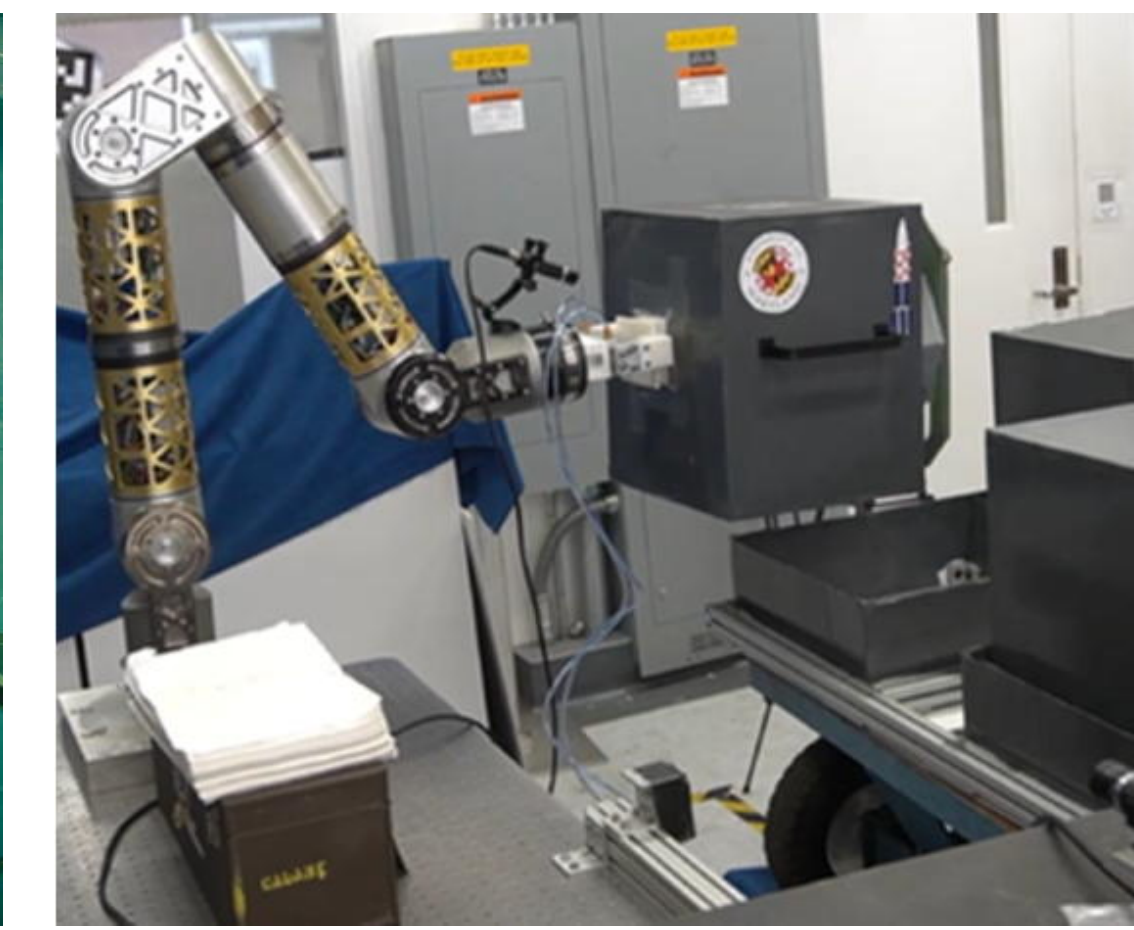
Space Utility Vehicle operations



Robotic rescue of EVA crew

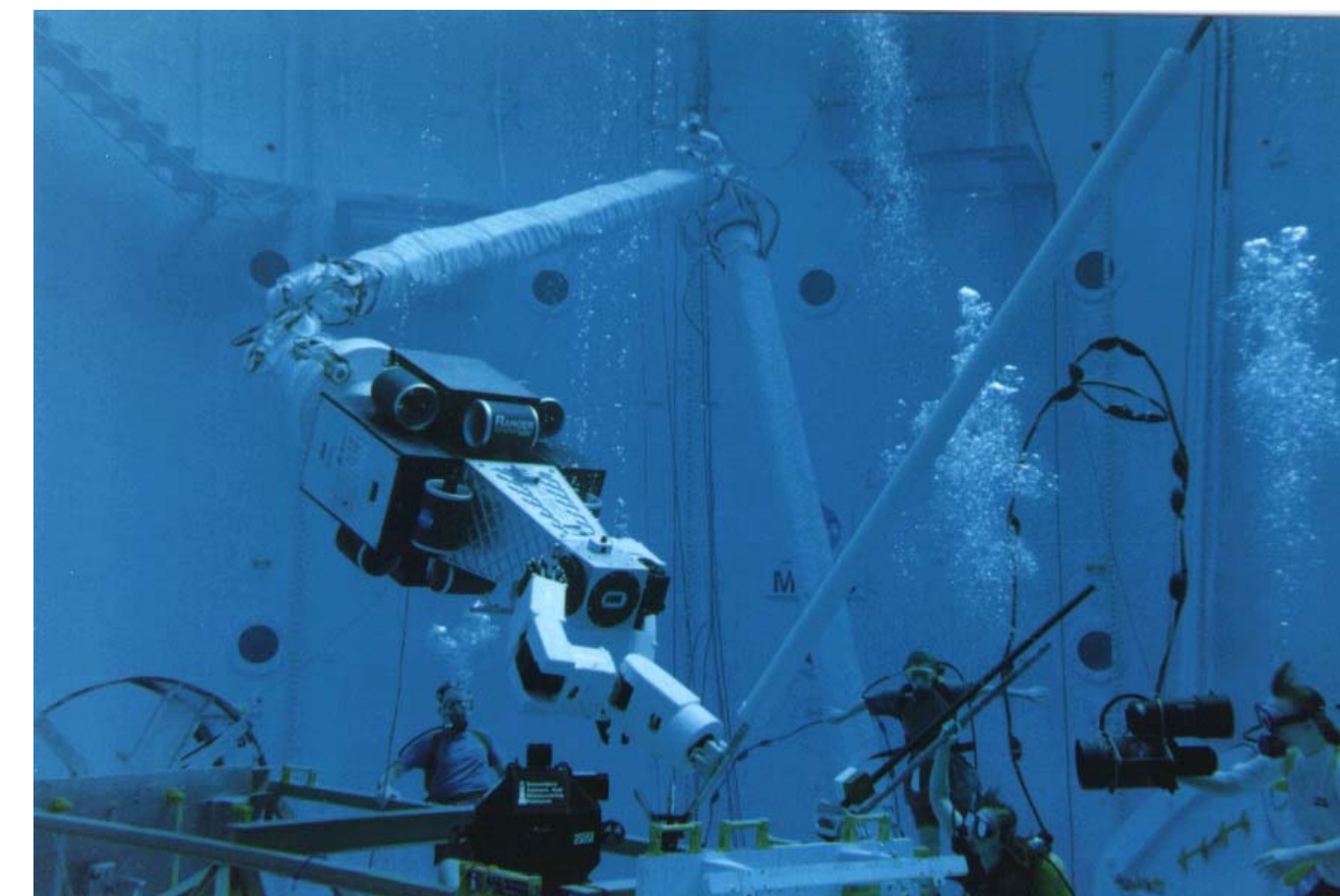


Suit-integrated manipulators

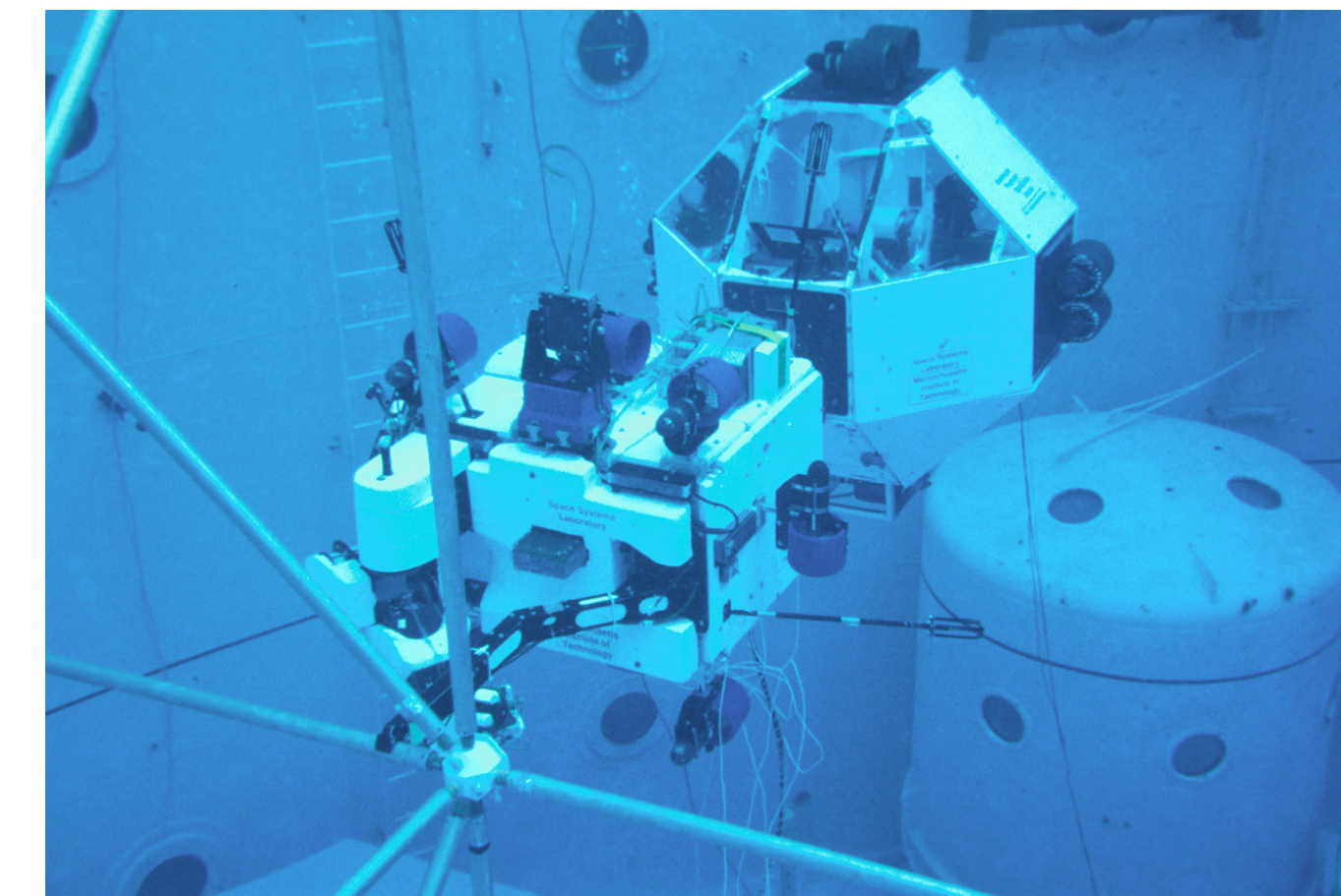


Lunar surface logistics

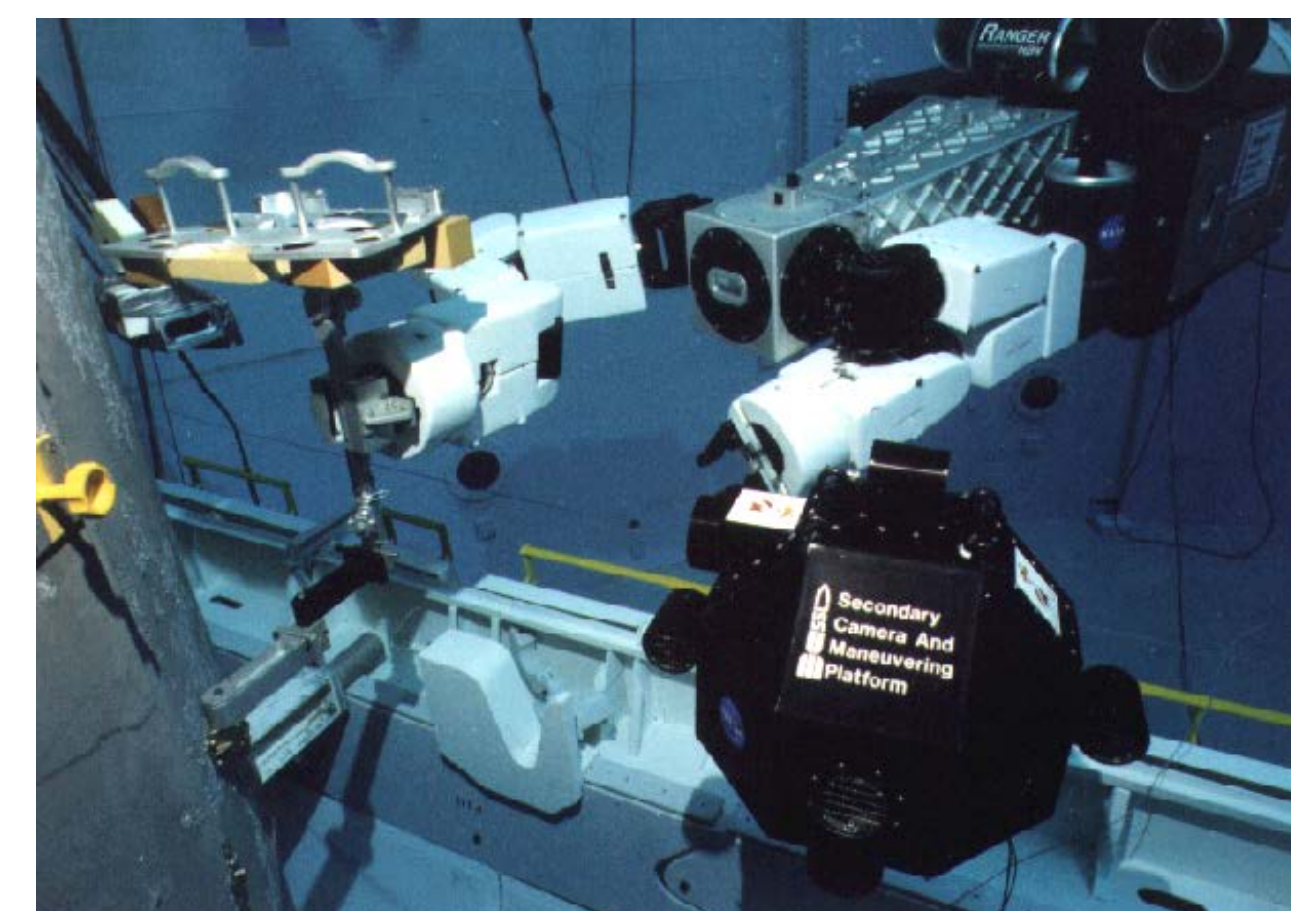
Multi-Agent Collaboration



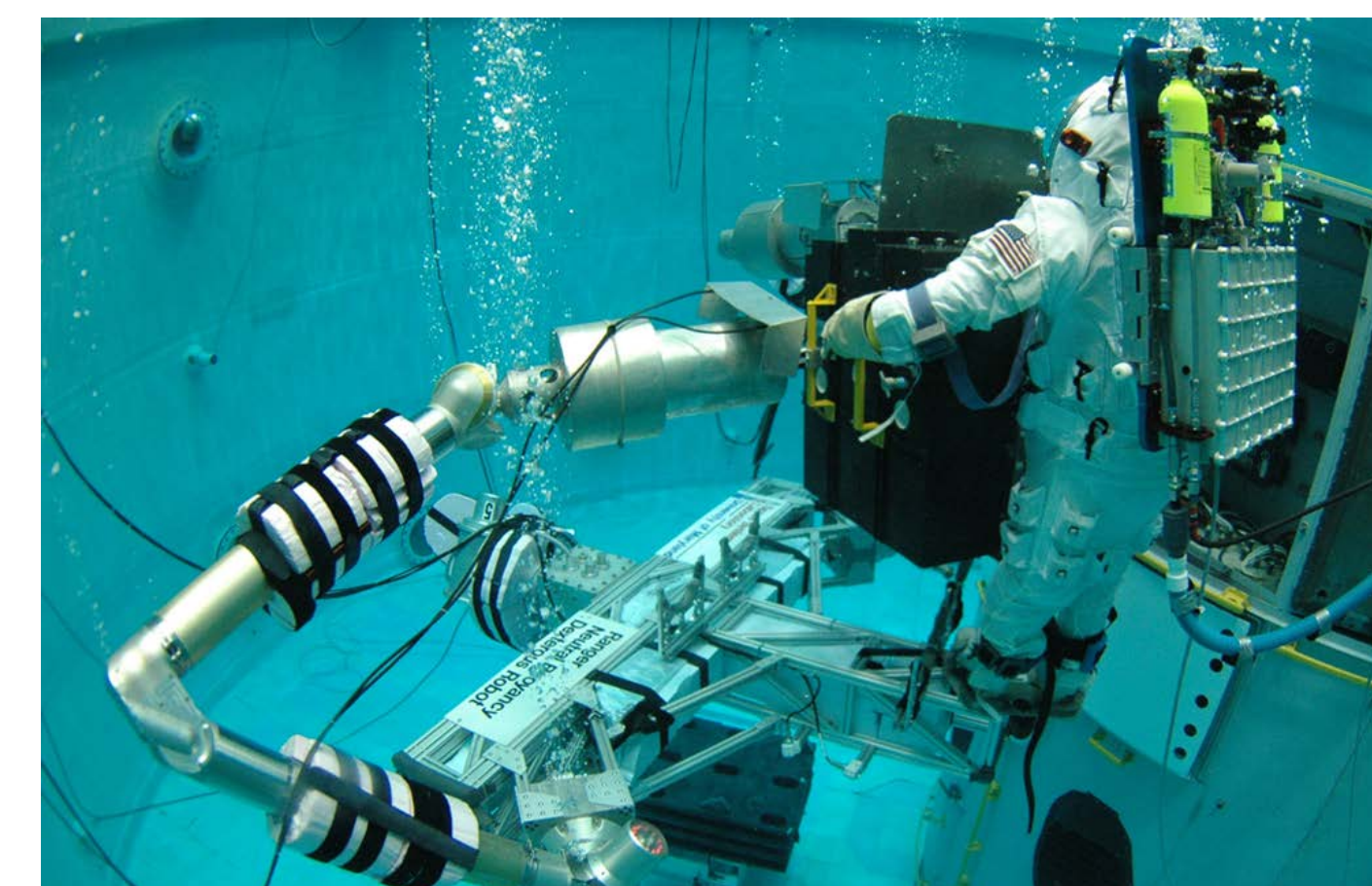
Dexterous robot/positioning arm



Dexterous robot/
orbital maneuvering vehicle



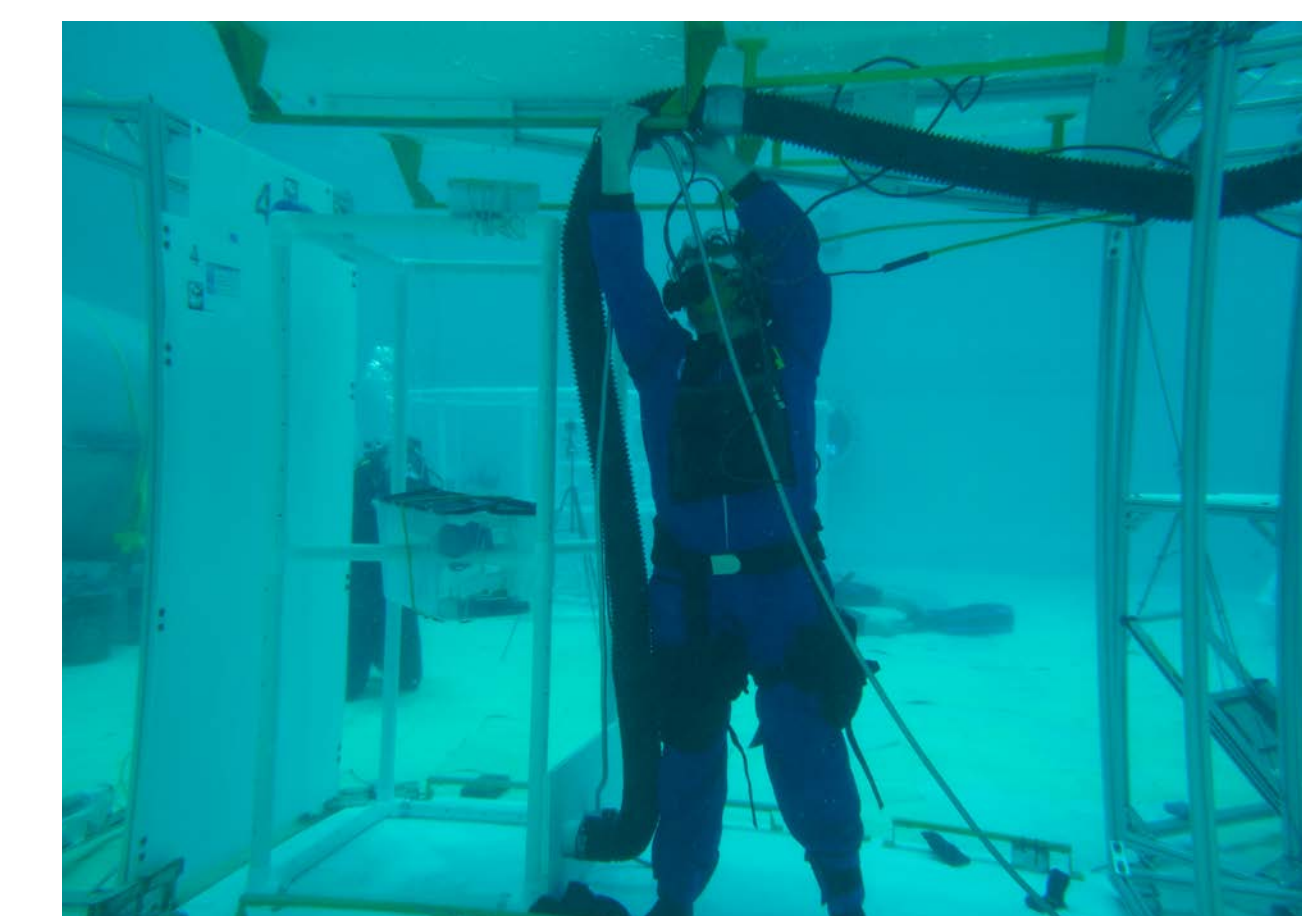
Site preparation using free-flying
external camera views



EVA/robotic direct collaboration



Planetary surface assembly
and maintenance



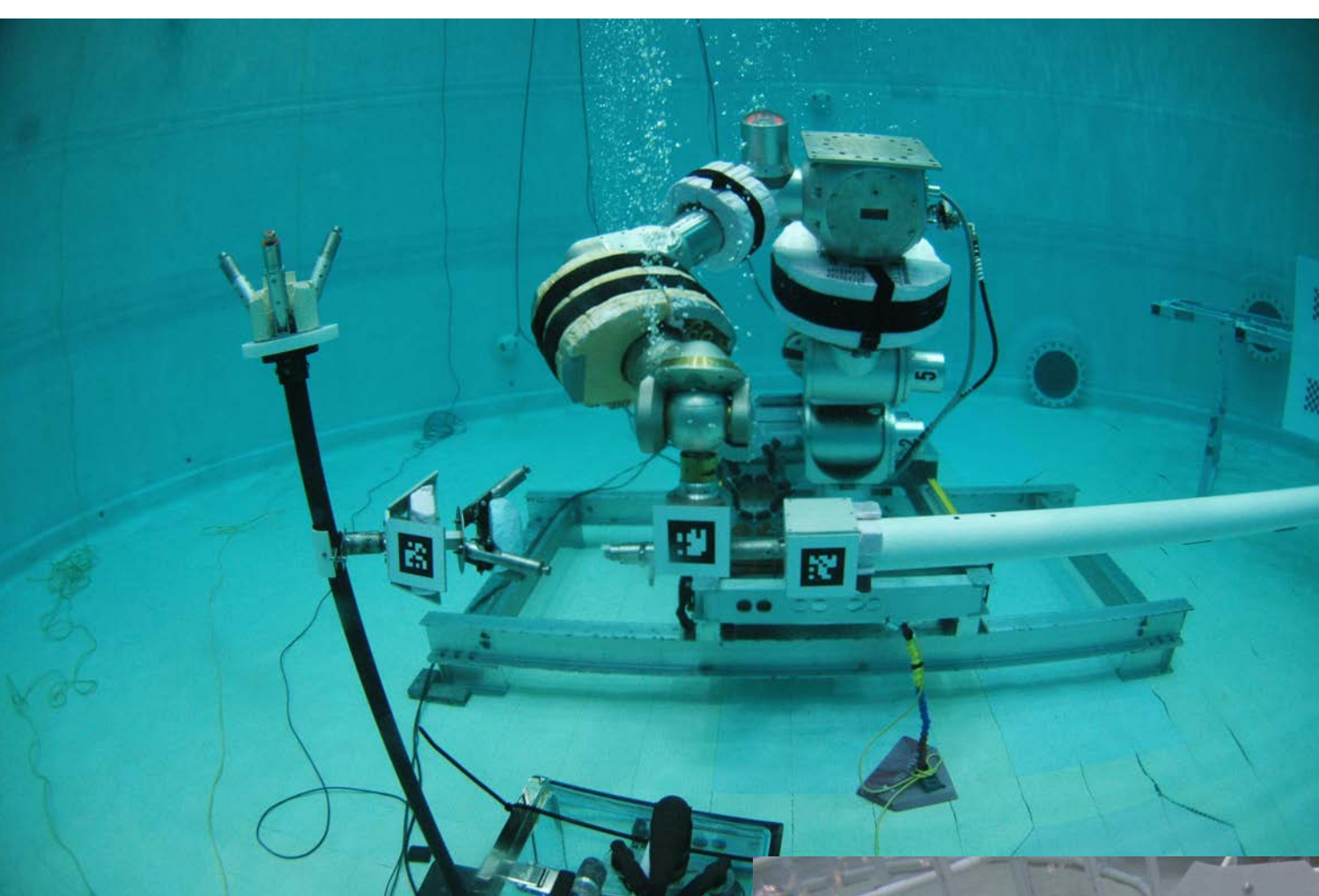
Interior outfitting of orbital habitats

For more than three decades, the University of Maryland Space Systems Laboratory has been focused on experimental investigation of in-space servicing, assembly, and maintenance using humans, robots, and human/robot or robot/robot collaborations. This has resulted in more than 200 publications. Systems developed to support this testing include seven rovers, eight dexterous manipulators, and six spacesuits or space suit simulators. The University of Maryland is home to the Neutral Buoyancy Research Facility, used for realistic representations of microgravity and planetary surface gravity levels. Those interested in more details are encouraged to contact the author.

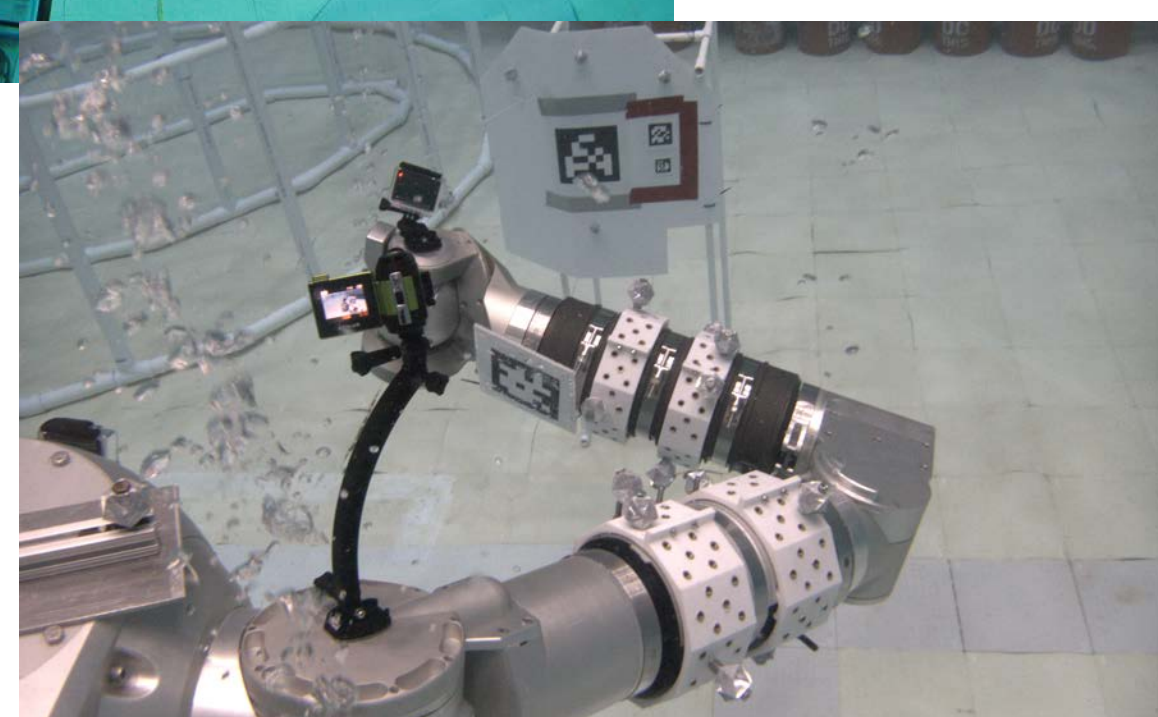
Space Systems Laboratory
University of Maryland



Autonomous Assembly



External vision-based
(above)/internal
vision-based (right)



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